

SIT723 Research Techniques & Applications

Task 2.2C: Identifying Research Themes and Narrative

1. Project Context

This document identifies the main research themes derived from the reviewed literature and constructs a narrative flow for the project “Measuring Semantic Stability of Interpretation in Large Language Models: A UMLS-Grounded Semantic Entropy Framework for Clinical NLP Evaluation.” The reviewed papers show that although large language models (LLMs) often achieve strong benchmark performance, they can still behave inconsistently when the same meaning is expressed through slightly different wording. This suggests that traditional evaluation metrics such as accuracy do not fully capture semantic stability.

Six themes were synthesised from the literature: prompt sensitivity, meaning-preserving perturbation and robustness, uncertainty and semantic entropy, biomedical grounding via UMLS, reliability and the evaluation illusion in medicine, and cross-domain QA validation. Together they support the narrative arc of the project: from input instability to structured measurement, to clinically grounded evaluation, and finally to cross-domain generalisation.

2. How the Themes Were Identified

The themes were derived through a combined deductive–inductive thematic analysis of the reviewed literature, following the approach commonly used in narrative and integrative literature reviews. The identification process had four explicit stages.

First, each paper was read in full and annotated along three deductive dimensions aligned with the research question: (i) the stage of the LLM pipeline it addresses (input, intermediate reasoning, or output), (ii) its evaluation target (accuracy, robustness, calibration, uncertainty, or grounding), and (iii) its domain (general NLP, biomedical NLP, or clinical decision-making). These dimensions were chosen because the project explicitly studies how meaning-preserving input changes propagate to output-level semantic instability in the clinical domain.

Second, an inductive open-coding pass assigned short descriptive codes to each paper’s principal claim (e.g., “formatting sensitivity”, “synonym-substitution attack”, “concept-level normalisation”, “uncertainty proxy”, “hallucination taxonomy”). Codes were then grouped by conceptual similarity and method family rather than by publication venue or date.

Third, code clusters were iteratively merged and split until each candidate theme met three retention criteria: (1) it was supported by at least three independent papers, (2) it was conceptually distinct from the other themes with minimal overlap, and (3) it made a direct and explicit contribution to answering the research question on semantic stability. Candidate clusters that failed any criterion (for example, a standalone “prompt engineering” cluster) were absorbed into broader themes.

Fourth, the resulting themes were ordered to form a coherent narrative that mirrors the logical flow of the project: Theme 1 establishes the phenomenon (LLMs are input-sensitive), Theme 2 provides

the methodological toolkit to expose that sensitivity, Theme 3 introduces the measurement framework (semantic entropy), Theme 4 grounds measurement in clinical semantics via UMLS, Theme 5 motivates the clinical stakes and the evaluation illusion, and Theme 6 situates the work within cross-domain QA validation.

Each theme therefore reflects both a substantive cluster in the literature and a necessary building block in the study’s argument.

3. Dataset Selection for the Study

Three publicly available datasets are used to evaluate semantic stability across complementary settings. The table below summarises their scope and role; it is retained in tabular form because the content is strictly comparative reference data rather than synthesised literature.

Dataset	Size / Type	Availability and Role in This Study
MedMentions ST21pv	4,392 PubMed abstracts with 350,000+ UMLS-linked mentions; ST21pv subset with 21 preferred semantic types. Primary dataset. Planned sample: n = 550.	Publicly available. Main dataset for concept-level semantic entropy because it provides UMLS CUI grounding.
SQuAD 2.0	100,000 answerable and 50,000 unanswerable questions. Secondary dataset. Planned answerable sample: n = 200.	Publicly available. Used for cross-domain validation of semantic stability beyond clinical NLP.
BioASQ Task B Phase A	Biomedical QA dataset. Planned sample: n = 150, including factoid and yes/no questions.	Publicly available. Used as an intermediate biomedical QA validation set between MedMentions and SQuAD 2.0.

4. Research Themes: Synthesised Narrative

The six themes are presented below as cohesive narrative sections. Within each theme, the reviewed papers are synthesised into a single argument rather than listed individually, and citations are given in IEEE numbered form. Full bibliographic details are provided in the References section at the end of this document.

4.1 Theme 1: Prompt Sensitivity and Semantic Consistency in Large Language Models

Guiding question: How do meaning-preserving changes in input reveal instability in LLM behaviour?

A consistent finding across the general-NLP literature is that large language models can produce markedly different outputs when the input is rephrased in ways that do not alter its meaning. Sclar et al. [5] demonstrate that purely cosmetic changes to prompt formatting, spacing, punctuation, or

option ordering can shift benchmark accuracy by tens of percentage points, even though the underlying task is unchanged. Zhuo et al. [6] generalise this observation in their ProSA framework by introducing PromptSensiScore, which quantifies how much a model’s output distribution shifts under controlled prompt variations and shows that sensitivity is not a model-wide constant but varies systematically across tasks and datasets.

Moving from surface formatting to meaning-level paraphrase, Raj et al. [8] show that LLMs frequently give different answers to semantically equivalent questions, establishing semantic inconsistency as a first-class reliability problem. This is complemented by the broader landscape review of Novikova et al. [24], who distinguish several facets of consistency, factual, hypothetical, compositional, and moral, and argue that each requires its own evaluation criterion; in their taxonomy, the present project sits within the factual and compositional strand.

A partial mitigation has emerged in the form of consistency-based decoding. Wang et al. [7] show that sampling multiple reasoning paths and taking the majority answer (self-consistency) materially improves accuracy, which implies that stable outputs are more likely to be correct and therefore that inconsistency is itself a signal of unreliability. Wei et al. [35] provide the methodological backdrop for this line of work through chain-of-thought prompting, which links output structure, intermediate reasoning, and stability.

Taken together, these papers establish the phenomenon that motivates the project: meaning-preserving input variation exposes instability that standard accuracy metrics hide. This theme therefore justifies the study’s first methodological commitment to evaluate LLMs under controlled paraphrastic variation rather than on single-prompt accuracy.

4.2 Theme 2: Adversarial Robustness and Meaning-Preserving Perturbations

Guiding question: Can semantic instability be exposed systematically through controlled perturbations?

If Theme 1 establishes that LLMs are input-sensitive, Theme 2 provides the methodological machinery to expose that sensitivity in a principled way. Goyal et al. [9] offer a comprehensive survey of adversarial attacks and defences in NLP, and crucially distinguish meaning-preserving perturbations (which must leave semantics intact) from meaning-changing ones (which alter the ground truth). This distinction is foundational for the present study, which deliberately restricts itself to the former so that any change in output can be attributed to the model rather than to the input.

Classical attack methods demonstrate how fragile NLP models can be under minimal, meaning-preserving edits. Jin et al. [25] introduce TextFooler, which uses contextual synonym substitution to flip classification decisions while preserving sentence meaning and grammaticality, and Ebrahimi et al. [26] show with HotFlip that even single-character or token-level swaps can substantially change predictions. These works justify the use of synonym- and paraphrase-based perturbations as principled probes of semantic stability rather than as mere noise.

More recent LLM-focused work confirms that scale alone does not resolve the problem. Alahmari et al. [10] evaluate several modern LLMs under perturbation and find that instability persists even in

very large models, and Xu et al. [11] go further by showing that an LLM can generate adversarial prompts capable of misleading itself or its peers, evidence that instability is partly intrinsic to the way these models represent language. Şahin [32] contributes a practical comparative study of text augmentation techniques (including back-translation and synonym replacement), directly informing the perturbation-generation pipeline used in this project.

Within the biomedical domain, Moradi and Samwald [36] show that adversarial training improves both accuracy and robustness in biomedical language models, providing a comparison point for discussing how robustness and stability relate in specialised settings. Collectively, these studies supply the perturbation toolkit, the meaning-preservation constraint, and the empirical precedent that underpin the experimental design of this study.

4.3 Theme 3: Uncertainty Estimation, Calibration, and Semantic Entropy

Guiding question: Why is semantic entropy a stronger evaluation framework than token-level confidence or plain accuracy?

Quantifying instability requires a principled measure, and the reviewed literature converges on the view that token-level probability is not that measure. Guo et al. [30] provide the foundational finding that modern neural networks are systematically overconfident, which means that raw softmax probabilities cannot be taken at face value as reliability estimates. Kadavath et al. [23] extend this critique to LLMs, showing that models only partially “know what they know”, and that internal confidence frequently fails to track correctness.

The theoretical pivot for this project is provided by Kuhn, Gal, and Farquhar [1], who formalise semantic entropy: an uncertainty measure computed over clusters of semantically equivalent generations rather than over token sequences. By collapsing surface variation and keeping only meaning-level variation, semantic entropy directly operationalises the notion of stability that Themes 1 and 2 argue is missing from standard evaluation. Farquhar et al. [2] then validate this framework at scale in Nature, showing that higher semantic entropy is strongly associated with hallucinated or incorrect outputs, which establishes semantic entropy as a practically useful reliability signal rather than only a theoretical construct.

Complementary measurement approaches are positioned around this core. Manakul et al. [40] propose SelfCheckGPT, a zero-resource consistency-based hallucination detector that samples multiple responses and measures their agreement without requiring an explicit semantic equivalence criterion; this provides a natural baseline against which UMLS-grounded semantic entropy can be compared. Savage et al. [20] study uncertainty proxies based on output similarity and calibration in medical tasks, showing that consistency-derived signals transfer into the clinical domain. Geifman and El-Yaniv [31] contribute selective classification, which uses uncertainty to decide when to abstain, giving a broader reliability-oriented evaluation paradigm that this study aligns with.

Read together, this theme motivates the central methodological choice of the project: instead of evaluating LLMs by accuracy alone, semantic entropy, grounded in a well-defined equivalence

criterion, is used as the primary stability metric, with calibration and consistency baselines providing comparative context.

4.4 Theme 4: Biomedical LLMs, UMLS Grounding, and Concept-Level Evaluation

Guiding question: Why is the medical domain especially suitable for semantic stability evaluation?

Semantic entropy requires a concrete criterion for deciding when two generations “mean the same thing”. In open-domain text this criterion is notoriously fuzzy, but in the biomedical domain a formal answer already exists. Bodenreider [38] describes the Unified Medical Language System (UMLS), whose Concept Unique Identifiers (CUIs) map many surface forms (e.g., “heart attack”, “myocardial infarction”, “MI”) to the same concept. This structure provides the formal semantic equivalence relation that general-domain semantic entropy otherwise lacks and is the main reason this study is anchored in clinical NLP.

The data and tooling to exploit this grounding are well established. Mohan and Li [28] introduce MedMentions, the main biomedical entity-linking corpus, which provides the annotated UMLS-grounded mentions used as the primary dataset here, and Neumann et al. [29] provide ScispaCy, a biomedical NLP toolkit with UMLS linking that supports the preprocessing and CUI-mapping pipeline. Together, they make concept-level evaluation operationally feasible.

Within biomedical language modelling, Lee et al. [14] establish BioBERT as a foundational domain-specific transformer, and Gu et al. [12] extend this line with PubMedBERT, showing that pretraining entirely within the biomedical domain outperforms mixed-domain pretraining. Tinn et al. [13] then study fine-tuning behaviour and stability specifically in biomedical NLP, suggesting that biomedical models can differ in stability as well as in accuracy, a finding directly relevant to this study’s comparative evaluation. At the large-model end, Labrak et al. [15] release BioMistral, a family of open-source medical LLMs that provides a contemporary domain-specific comparator.

Two recent works tie UMLS grounding directly to LLM behaviour. Afshar et al. [18] show that UMLS-aware grounding improves diagnosis-generation quality and evaluation alignment, supporting the use of CUI-level clustering in this project. Dobbins [19] reports that, despite their general strength, LLMs still struggle with direct biomedical concept normalisation, which reinforces the case for measuring instability at the concept level rather than the token level. This theme therefore provides both the rationale and the practical infrastructure for UMLS-grounded semantic entropy.

4.5 Theme 5: Reliability, Hallucination, and the Evaluation Illusion in Medicine

Guiding question: Why does this problem matter in practice?

Semantic instability is not an abstract measurement concern; in medicine it translates directly into clinical risk. Ji et al. [3] provides the foundational survey of hallucination in natural language generation, classifying its types and causes across NLG systems, and Huang et al. [4] update this taxonomy specifically for LLMs, showing that hallucination and semantic drift remain central reliability issues even for the most recent model generations. These surveys frame semantic instability as a practical rather than theoretical problem.

The gap between benchmark success and clinical reliability is made explicit by three complementary studies. Singhal et al. [16] demonstrate that LLMs can encode clinical knowledge to a remarkable degree and reach strong medical benchmark performance, but Hager et al. [17] show in *Nature Medicine* that this benchmark strength does not translate into reliable clinical decision-making on realistic tasks. Agrawal et al. [21] crystallise this critique as the “evaluation illusion” of LLMs in medicine, arguing that current evaluation practices systematically overstate clinical readiness. This triad directly motivates the significance of the present study: a model that appears accurate on a benchmark but that is semantically unstable under meaning-preserving variation is exactly the kind of system the evaluation illusion warns about.

Constructive responses in the same body of work point toward uncertainty-aware evaluation. Zhou et al. [22] show that uncertainty-aware modelling can reduce misdiagnosis in explainable disease diagnosis, reinforcing the practical value of better uncertainty estimation, and Thirunavukarasu et al. [33] provide a broader review of LLM uses and risks in medicine that situates these reliability concerns at the domain level.

Finally, two works help scope the project. Boag et al. [34] show that clinical text carries predictive information not captured by structured codes alone, which justifies evaluating LLMs on free-text clinical language rather than only on code-based tasks, while Névél et al. [37] review the multilingual limitations of clinical NLP, which the present study uses to acknowledge an explicit scope boundary: its findings apply to English clinical NLP.

4.6 Theme 6: Cross-Domain QA Context and Narrative Bridging

Guiding question: How does QA context support the move from clinical concept normalisation to broader validation?

To test whether the proposed stability framework is domain-specific or more generally useful, the study extends evaluation into open-domain question answering. Rajpurkar et al. [27] introduce SQuAD 2.0 by combining answerable and unanswerable questions, which makes it a particularly informative cross-domain benchmark: a semantically stable model should not only give consistent answers to paraphrased answerable questions, but should also consistently abstain on unanswerable ones. Zhu et al. [39] provide a comprehensive survey of open-domain QA systems and benchmarks, establishing QA as a standard setting for studying semantic and epistemic uncertainty and clarifying where SQuAD 2.0 sits within that landscape.

This theme therefore serves a bridging role in the overall narrative: it connects the UMLS-grounded clinical evaluation of Themes 3–5 with a general-domain validation setting, which strengthens any claim that semantic entropy captures a genuine property of LLM behaviour rather than an artefact of one dataset or one domain.

5. Narrative Flow Across the Themes

The six themes are not independent. They form a single argument that drives the research design of the project.

- Theme 1 establishes that LLMs are sensitive to meaning-preserving input changes and that accuracy-based evaluation hides this.
- Theme 2 contributes the methodological instruments, meaning-preserving perturbation and adversarial robustness testing, needed to expose that sensitivity systematically.
- Theme 3 supplies the measurement framework, semantic entropy, that converts observed inconsistency into a principled uncertainty signal.
- Theme 4 grounds that measurement in clinical semantics via UMLS, which is what makes concept-level semantic entropy well-defined and reproducible.
- Theme 5 situates the work in the clinical reliability literature and the “evaluation illusion” critique, explaining why semantic stability matters in practice.
- Theme 6 broadens the evidence base through cross-domain QA validation, so the framework’s usefulness is not confined to a single corpus.

Read as a single narrative, the themes move from phenomenon, to method, to metric, to grounding, to clinical significance, to cross-domain generalisation. This is the structure the project follows and the contribution it is positioned to make: a UMLS-grounded semantic entropy framework that measures semantic stability of LLM interpretations in clinical NLP and validates its usefulness beyond a single domain.

6. References

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